



實習結果報告

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Internship Provider



實習機構

- 利優生醫科技股份有限公司：產品開發部

服務項目

- 霧化給藥技術：NEBreath、AniFree(動物)、BREEZONE

實習內容

- 霧化片顆粒性能測試與分析、個人專題：全橋直流馬達換向與控速

Responsibilities

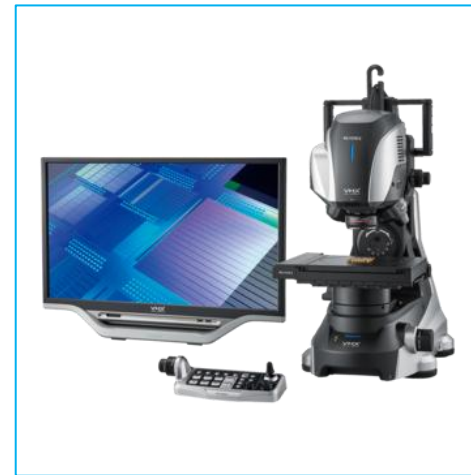
- 霧化片性能測試：比較不同間公司霧化片的差別與回報
- 使用儀器：Malvern Spraytec、KEYENCE VHX-X1 數位顯微鏡



Sample設定

校正

測量：
顆粒直徑
GSD
Dv(50)



測量前
表面觀察

Spraytec

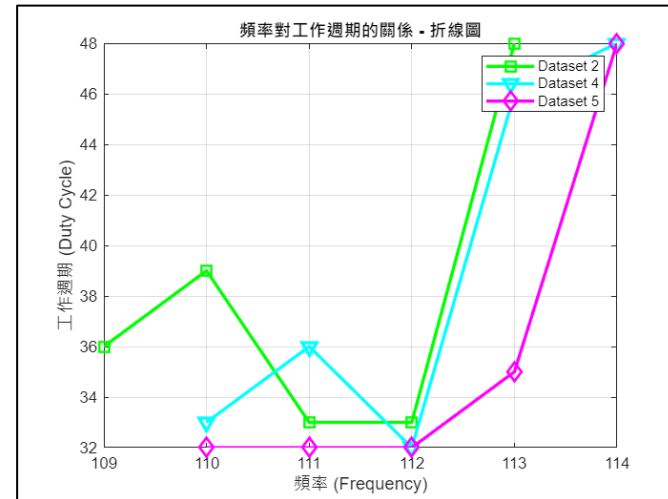
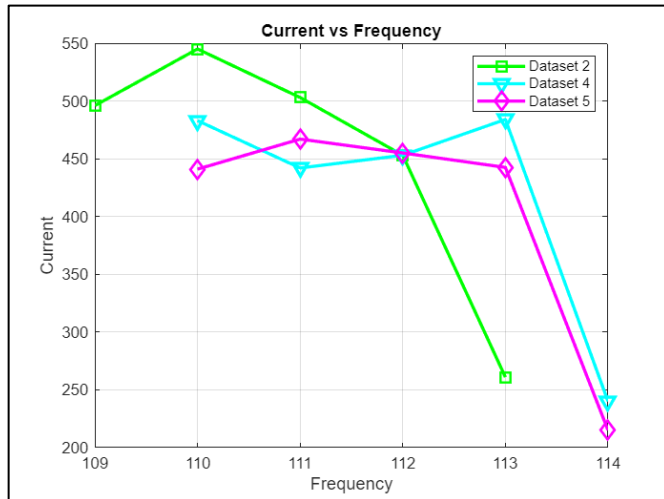
再次測量對
比

實驗記錄：每個廠家重複測10片，彙整資料

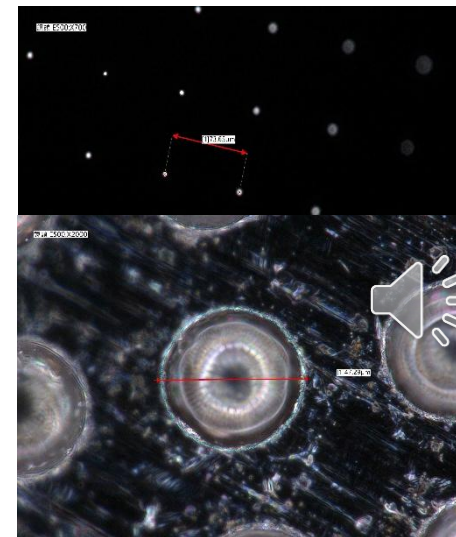
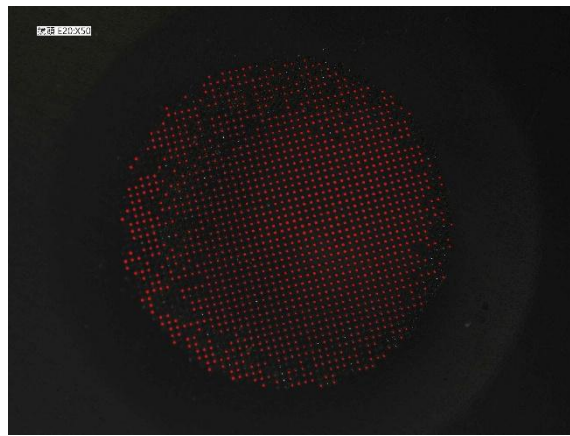
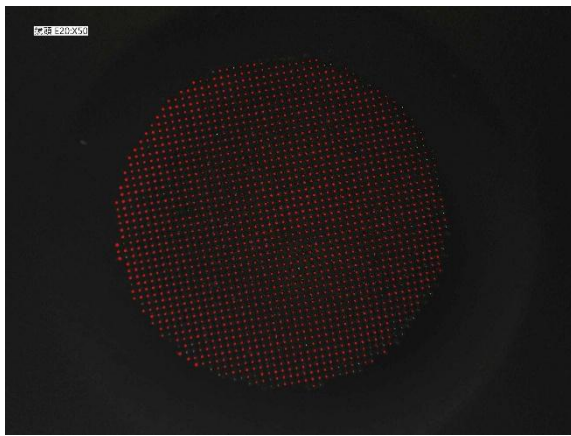
Duty	Freq	Count	Spray	<Sum(%)	GSD	Dv(50)	Concentration	Duty	Freq	Count	Spray	<Sum(%)	GSD	Dv(50)	
30%		108	439.5-443	可	48.21%	1.853	5.126	30%	29%	101		55.77%	1.59	4.641	
				可	47.01%	1.639	5.212	30%	29%	100		55.15%	1.785	4.681	(噴一段時間)
30%	101	480		可	47.38%	1.645	5.187	30%	29%	99		61.90%	1.665	4.24	
29%	101	429.5		可	48.47%	1.633	5.105	30%	29%	102		60.39%	1.689	4.308	(再調前一段距離) / (噴完)
								30%	29%	103		60.67%	1.622	4.34	
								30%	29%	98		66.64%	1.58	4.041	(spraytec最小值5)
								30%	29%	104		63.76%	1.595	4.175	
								30%	29%	109		58.71%	1.715	4.421	
												52.07%	1.799	4.859	(噴完，電流掉到280)
								40%	29%	101					
								40%	31%	101		71.71%	2.177	3.383	(不穩)
								40%	31%	100		74.81%	1.701	3.51	(噴一段時間)
								40%	31%	99		60.56%	1.53	4.416	
								40%	31%	98					(重補)

Responsibilities

- 用MATLAB繪製關係圖(取濃度30%~60%甘油)

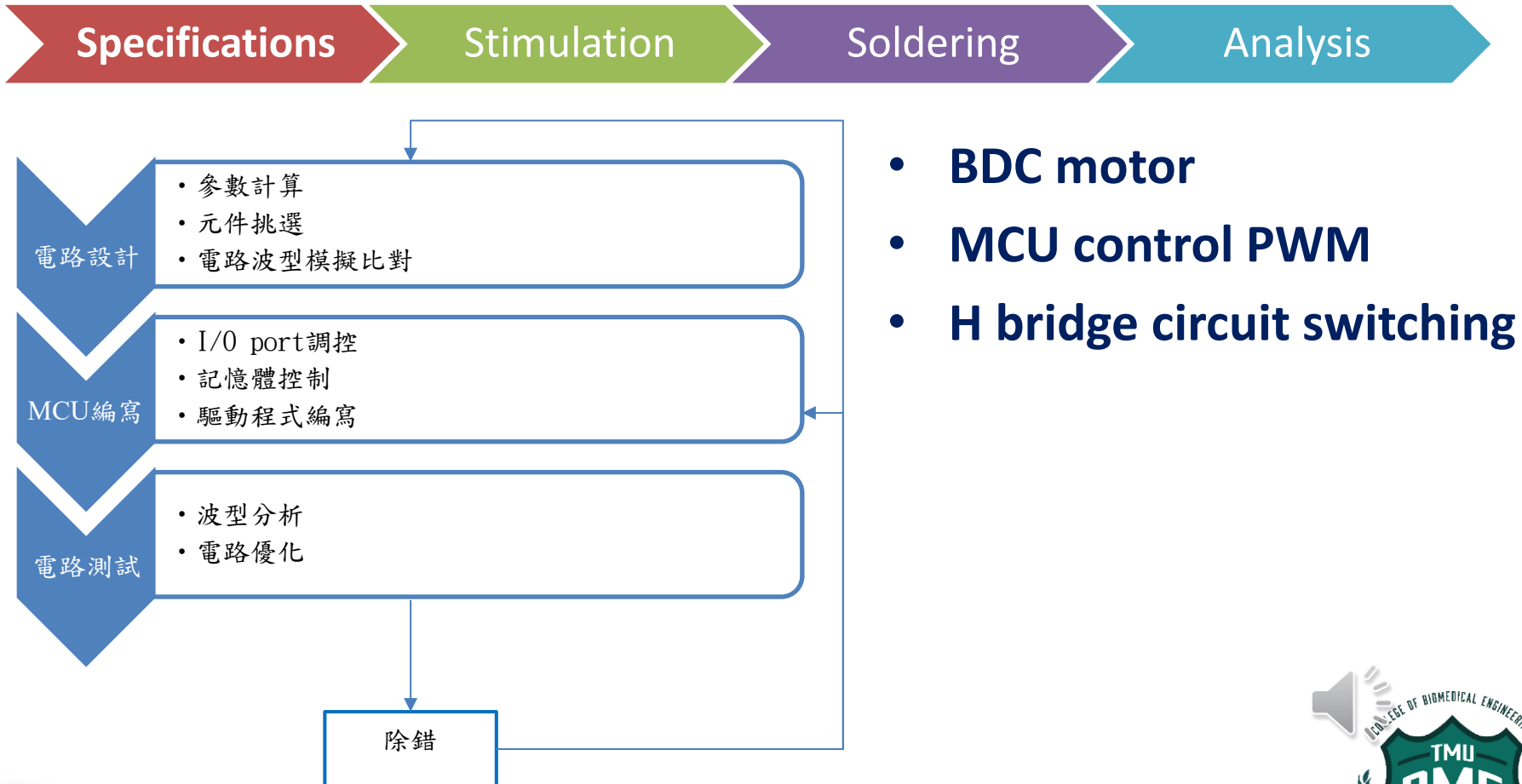


- 噴霧測試前後孔徑變化



Hands-on experience

Personal Project:

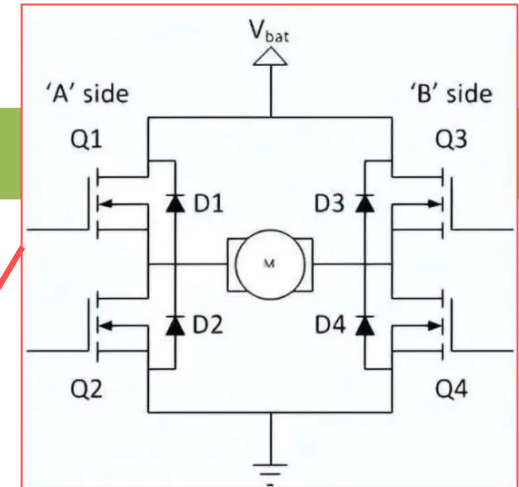
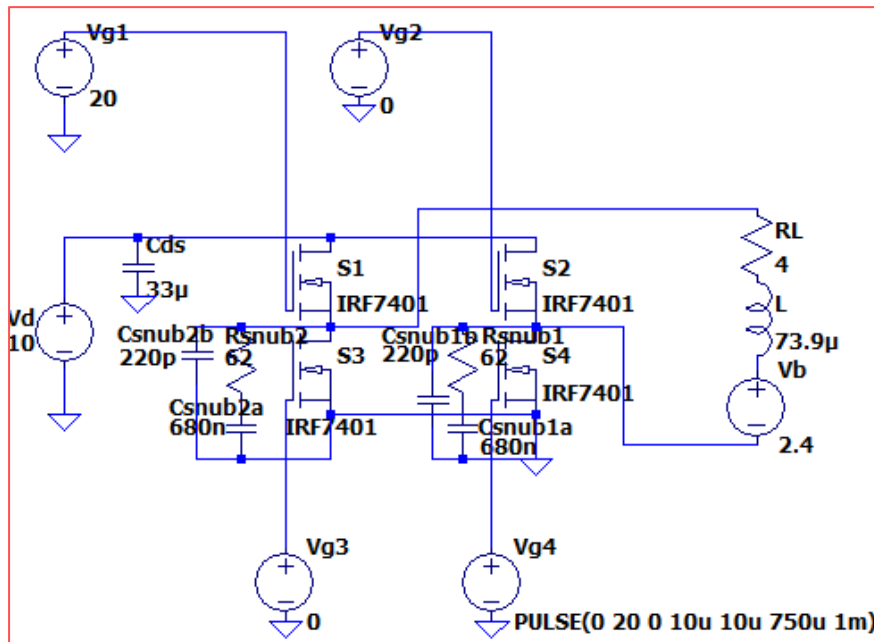


Hands-on experience

- Spice model : H-bridge motor driver

Stimulation

- Spice model : H-bridge motor driver



$$V = IR + L \frac{di}{dt} + V_b$$

$\downarrow$ $\downarrow$ $\downarrow$
 4.75Ω 13.9μ $\text{duty } 75\%$

$1.186 \rightarrow 1.2$

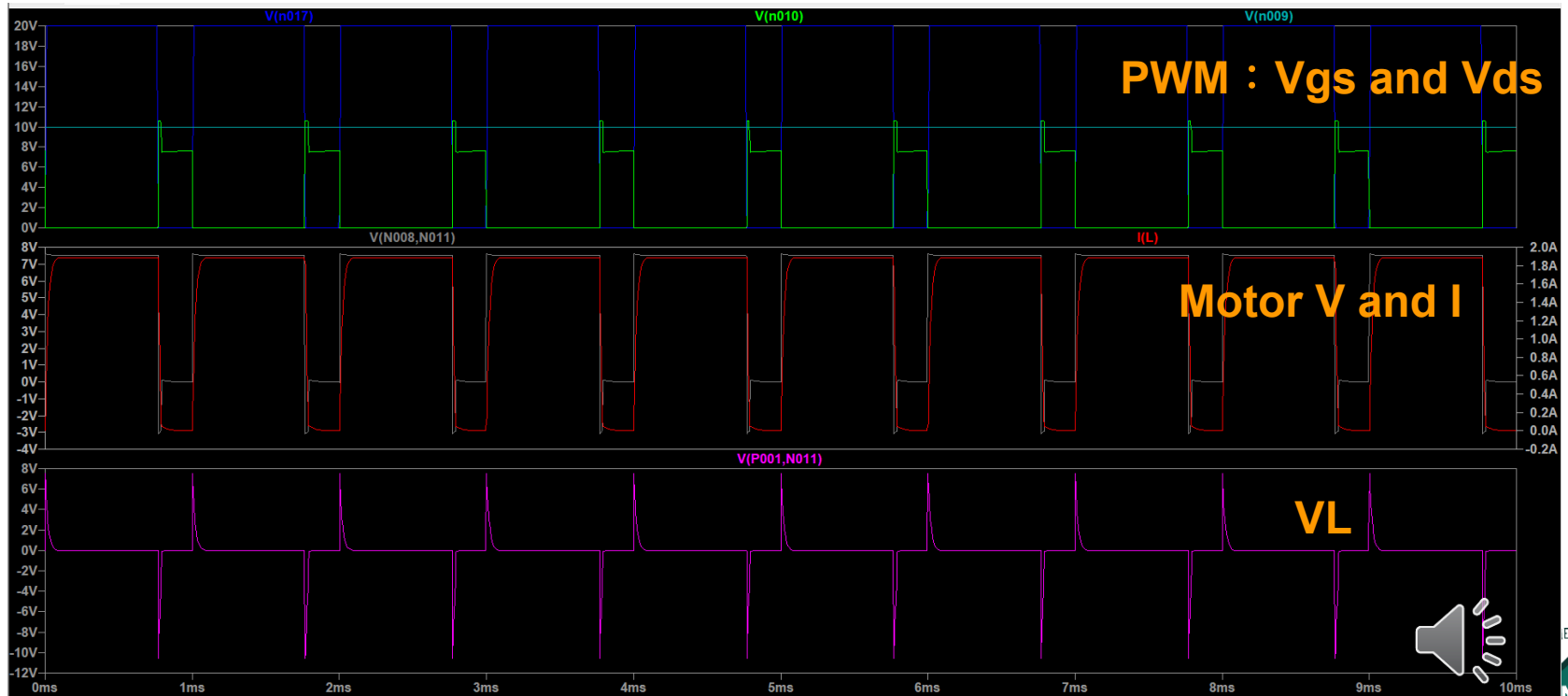
$$1.5 = 1.2 \times 4.75 + 13.9\mu \times \frac{1.2}{0.15\text{ms}} = V_b = 2.34\text{V}$$

$$I = \frac{V}{R} = \frac{1.5 - V_b}{\sqrt{4.75^2 + f \times 2\pi \times 1k \times 13.9\mu}} = 1.2\text{A}$$

Hands-on experience

- LTspice:

Stimulation

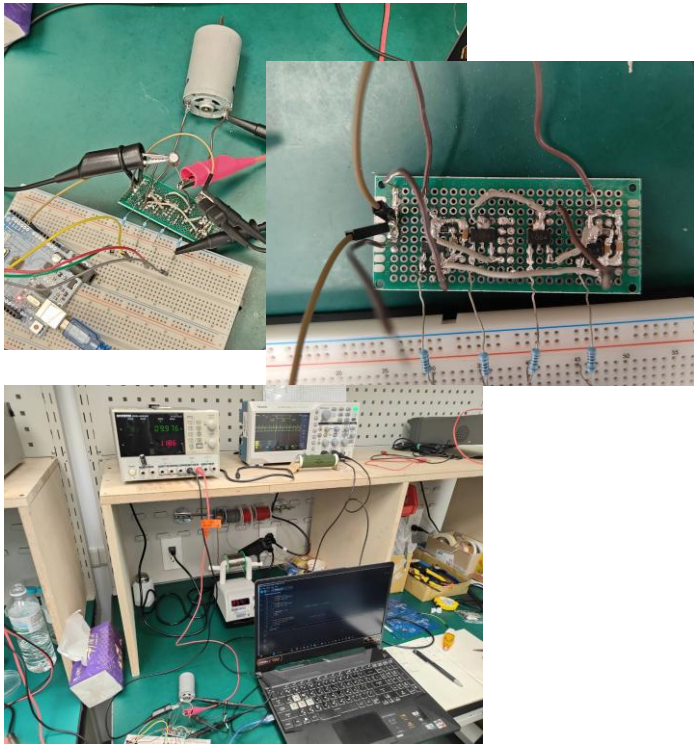


Hands-on experience

Personal Project:

Soldering

Analysis



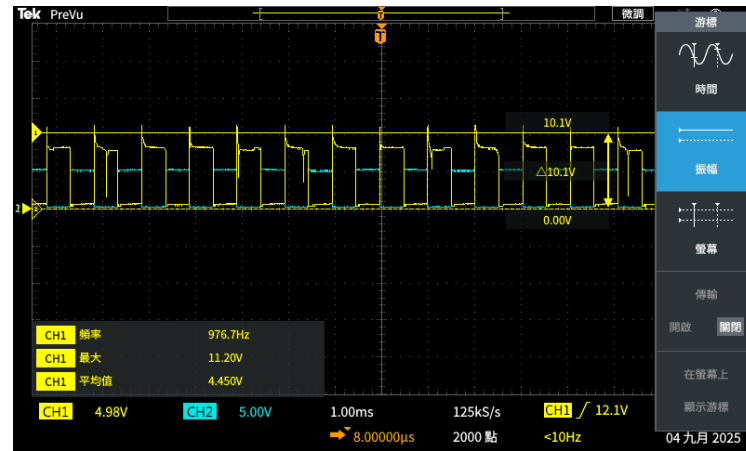
啟動電流太大，緩啟動與關閉保護電路

```
14
15 void loop() {
16     for (int i = 100; i>0; i-- ){
17         if (Confirm == 1){
18             analogWrite(5, (duty*i/100*25));
19             analogWrite(6, 0);
20             delay(100);
21         }
22         if (Confirm
23             analogWri
24             analogWri
25             delay(100
26         }
27     }
28
29     digitalWrite(
30     digitalWrite(
31     //digitalWrit
32     //digitalWrit
33     delay(500);

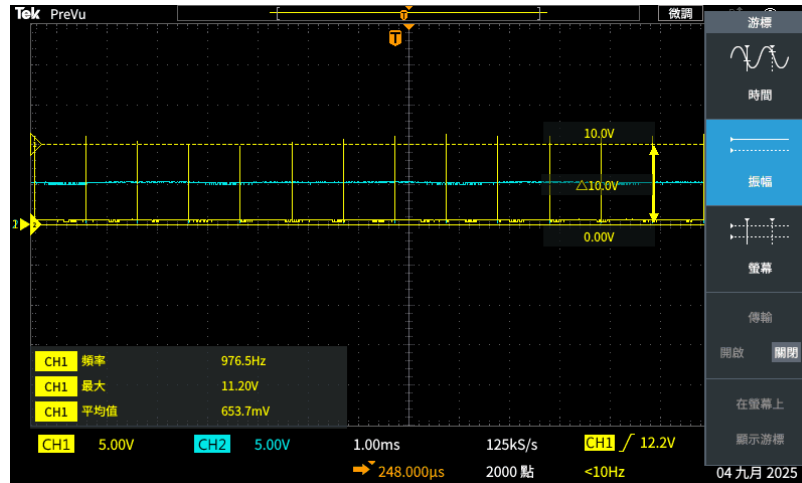
42 while (true){
43     Serial.println("\n=== PWM 控制系统 ==="); // 選擇腳位
44     Serial.println("選擇轉向");
45     int confirm = inputInt("順轉1 / 反轉0");
46     Confirm = confirm; // Confirm為全域變數避免迴圈結束失去數值
47
48     if (confirm == 1){ // 選擇1順轉，0反轉
49         digitalWrite(12, LOW);
50         digitalWrite(13, HIGH);
51         break;
52     }
53     if (confirm == 0){
54         digitalWrite(13, LOW);
55         digitalWrite(12, HIGH);
56         break;
57     }
58 }
59
60 for (int i = 0; i<255; i++){ // PWM緩啟動
61     int breakpoint = ChooseDirect(); // 依據方才的輸入決定PWM控制的MOSFET
62
63     if (confirm == 1){
64         analogWrite(5, (duty*i));
65         analogWrite(6, 0);
66     }
67     if (confirm == 0){
68         analogWrite(6, (duty*i));
69         analogWrite(5, 0);
70     }
}
```



- **50%**



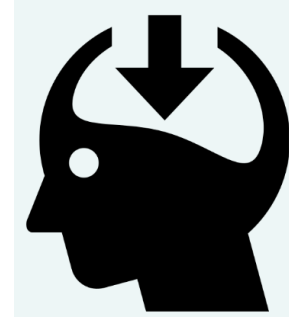
- **100%**



Summary

- 改進處：
執行速度
專業能力
訊息、眼界

學到了什麼



自學能力

電路設計基本流程
實事求是的態度

跨領域接觸

